

Mathematics for Computer Science

COMP1021: Durham University

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Part I

Calculus

1 Introduction to Calculus

1.1 Sets

Definition 1.1.1: A set is an unordered collection of distinct elements denoted as $A = \{x | \text{condition}\}$.

Some common set operations include:

- **Definition 1.1.2:** The union of two sets, $A \cup B$, includes all the elements of both sets.
- **Definition 1.1.3:** The intersection of two sets, $A \cap B$, includes any elements which are present in both sets.
- **Definition 1.1.4:** If A and B are finite sets, then the principle of inclusion-exclusion states that

$$A \cup B = A + B - (A \cap B) \quad (1)$$

- **Definition 1.1.5:** The set A is a subset of B , $A \subseteq B$, if every element of A is an element of B . A is a proper subset of B , $A \subset B$, if A is a subset of B and there is at least one element of B not in A .
- **Definition 1.1.6:** The powerset of a set A is the set of all subsets of A .

$$\mathcal{P}(A) = \{A' | A' \subseteq A\} \quad (2)$$

- **Definition 1.1.7:** The cartesian product of two sets, $A \times B$, is the set of all possible ordered pairs with the first element from A and the second from B .

Definition 1.1.8: For a finite set, the cardinality is the number of distinct elements that it contains.

1.2 Functions

For a function $f : A \rightarrow B$,

- **Definition 1.2.1:** A is known as the domain of f .
- **Definition 1.2.2:** B is known as the codomain of f .
- **Definition 1.2.3:** If $f(a) = b$, then b is the image of a .
- **Definition 1.2.4:** The pre-image of b is the set $\{a | f(a) = b\}$.
- **Definition 1.2.5:** f forms an injection if every element of A maps to a unique element in B , even if some elements of B don't have a mapping.
- **Definition 1.2.6:** f forms a surjection if every element of B maps to at least one element of A .
- **Definition 1.2.7:** f forms a bijection if it is both injective and surjective.

Definition 1.2.8: If for an unbounded set A , there is a bijective function $f : A \rightarrow \mathbb{N}$, then we say A is countably infinite, otherwise A is known as uncountably infinite.

1.3 Hyperbolic Functions

Definition 1.3.1: Given the unit hyperbola $x^2 - y^2 = 1$, the hyperbolic functions are defined by the coordinates of a point at which area $\frac{1}{2}\alpha$ has been swept out.

$$\begin{aligned} \sinh(\alpha) &= \frac{e^\alpha - e^{-\alpha}}{2} & \cosh(\alpha) &= \frac{e^\alpha + e^{-\alpha}}{2} & \tanh(\alpha) &= \frac{e^\alpha - e^{-\alpha}}{e^\alpha + e^{-\alpha}} & \sigma(\alpha) &= \frac{1}{1 + e^{-\alpha}} \\ \operatorname{sech}(\alpha) &= \frac{2}{e^\alpha + e^{-\alpha}} & \operatorname{cosech}(\alpha) &= \frac{2}{e^\alpha - e^{-\alpha}} & \coth(\alpha) &= \frac{e^\alpha + e^{-\alpha}}{e^\alpha - e^{-\alpha}} \end{aligned}$$

These functions can then be differentiated as follows:

$$\begin{aligned} \frac{d}{dx} [\sinh(\alpha)] &= \cosh(\alpha) & \frac{d}{dx} [\cosh(\alpha)] &= \sinh(\alpha) \\ \frac{d}{dx} [\tanh(\alpha)] &= \operatorname{sech}^2(\alpha) & \frac{d}{dx} [\sigma(\alpha)] &= \sigma(\alpha)(1 - \sigma(\alpha)) \end{aligned}$$

2 Multivariate Functions

2.1 Multivariate Derivatives

Definition 2.1.1: For a function $f(x, y)$, if we take $f = f_0$ to be fixed but unknown, then we can differentiate f as a function of x at $x = x_0$, giving us the partial derivative of f with respect to x which gives the gradient when moving in the x -direction from any point.

$$\frac{\delta f}{\delta x} [f(x, y)] = \lim_{x \rightarrow x_0} \left(\frac{f(x, y_0) - f(x_0, y_0)}{x - x_0} \right) = f_x \quad (3)$$

Definition 2.1.2: The vector of partial derivatives, denoted as ∇f , contains all the partial derivatives of a function. This forms a vector pointing in the direction of the greatest rate of increase of f , where its magnitude is the rate of increase in that direction. For example, for a function $f(x, y)$,

$$\nabla f = \begin{pmatrix} f_x \\ f_y \end{pmatrix} \quad (4)$$

Definition 2.1.3: The directional derivative of the function $f(x, y)$ gives the gradient at any point whilst looking in the direction given by the unit vector $\mathbf{v} = \begin{pmatrix} v_x \\ v_y \end{pmatrix}$.

$$\nabla_{\mathbf{v}} f = \lim_{h \rightarrow 0} \frac{f(x_0 + h\mathbf{v}) - f(x_0)}{h} \quad (5)$$

If the slope is approximately flat at a small scale, then the slope can be determined by

$$\nabla_{\mathbf{v}} f = \mathbf{v} \cdot \nabla f \quad (6)$$

From this, we get that, for a function $f(\mathbf{x}_0)$ and a direction $\mathbf{v}$,

$$f(\mathbf{x}_0 + h\mathbf{v}) \approx f(\mathbf{x}_0) + h\mathbf{v} \cdot \nabla f(\mathbf{x}_0) \quad (7)$$

2.2 Multivariate Extrema

Definition 2.2.1: A function that is continuous is in class C^0 , and a function that is continuous and has a continuous derivative is in class C^1 . A function f such that f' , f'' , ..., $f^{(k)}$ exist and are continuous is in class C^k .

Definition 2.2.2: A function f such that derivatives of all orders exist and are continuous is in class C^∞ is also smooth.

Definition 2.2.3: All the second-order partial derivatives of a function $f(x, y)$ can be gathered into a Hessian matrix.

$$\mathbf{H}_f = \begin{bmatrix} f_{xx} & f_{xy} \\ f_{yx} & f_{yy} \end{bmatrix} \quad (8)$$

This can be used to determine the form of bivariate extrema. Suppose $f(x, y) \in C^2$ and $f_x(x_0, y_0) = 0 = f_y(x_0, y_0)$ then,

- (x_0, y_0) is a local maximum if $\det(\mathbf{H}_f) > 0$ and $f_{xx} < 0$ at (x_0, y_0) .
- (x_0, y_0) is a local minimum if $\det(\mathbf{H}_f) > 0$ and $f_{xx} > 0$ at (x_0, y_0) .
- (x_0, y_0) is a saddle point if $\det(\mathbf{H}_f) < 0$.
- If $\det(\mathbf{H}_f) = 0$, then the test is inconclusive.

At a stationary point x_0 , the quadratic approximation of a function $f(x)$ is

$$f(x_0 + \mathbf{v}) \approx f(x_0) + \frac{1}{2} \mathbf{v}^T \mathbf{H}_f(x_0) \mathbf{v} \quad (9)$$

But as $\mathbf{H}_f(x_0)$ is real and symmetric, it has orthogonal eigenvectors $\mathbf{e}_1, \mathbf{e}_2, \dots, \mathbf{e}_n$ with eigenvalues $\lambda_1, \lambda_2, \dots, \lambda_n$, so we can write

$$\mathbf{v} = c_1 \mathbf{e}_1 + c_2 \mathbf{e}_2 + \dots + c_n \mathbf{e}_n \quad (10)$$

When substituted back into the original equation, this gives that

$$f(x_0 + \mathbf{v}) \approx f(x_0) + \frac{1}{2} (\lambda_1(c_1)^2 + \lambda_2(c_2)^2 + \dots + \lambda_n(c_n)^2) \quad (11)$$

Given this, we can use the eigenvalues to determine the nature of the stationary point x_0 ,

- If all eigenvalues are positive, then the stationary point is a local minimum.
- If all eigenvalues are negative, then the stationary point is a local maximum.
- If there is a mixture of positive and negative eigenvalues, then the stationary point is a saddle point.

2.3 Vector-Valued Functions

Definition 2.3.1: A vector-valued function takes an n -dimensional vector as input, and outputs an m -dimensional vector, e.g., $f : \mathbb{R}^n \rightarrow \mathbb{R}^m$.

Definition 2.3.2: The Jacobian matrix of a vector-valued function $f : \mathbb{R}^n \rightarrow \mathbb{R}^m$ can be formed by setting $\mathbf{J}_{i,j} = \frac{\delta f_i}{\delta x_j}$ to form,

$$\mathbf{J}_f = \begin{pmatrix} \frac{\delta f_1}{\delta x_1} & \dots & \frac{\delta f_1}{\delta x_n} \\ \vdots & \ddots & \vdots \\ \frac{\delta f_m}{\delta x_1} & \dots & \frac{\delta f_m}{\delta x_n} \end{pmatrix} \quad (12)$$

For a multivariate function $f : \mathbb{R}^n \rightarrow \mathbb{R}^m$, the Jacobian matrix can be used to linearly approximate f at x ,

$$f(x + h\mathbf{v}) \approx f(x) + \mathbf{J}_f(x)(h\mathbf{v}) \quad (13)$$

Part II

Linear Algebra

Part III

References

3 Module Professors

The below is accurate as of the 2025/26 academic year and will not be updated in subsequent years.

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4 Cited Works